

KUNAL PATIL

Senior Solution Architect | Principal Engineer | GenAI Architect

Pune, India | +91-9423859575 | patilkunals@gmail.com | linkedin.com/in/patilkunals

PROFESSIONAL SUMMARY

Senior technology leader and solution architect with 20+ years of experience across AI/GenAI, cloud architecture, and enterprise software delivery in Core Banking, Insurance, Retail, and Telecom. Proven record designing and delivering AI-powered, secure, scalable cloud-native architectures on AWS, integrating core banking with digital channels, payments, AML, CRM, and regulatory systems. Deep expertise in GenAI and Agentic AI solutions using RAG, LangChain, LangGraph, Model Context Protocol (MCP), and AWS Bedrock, combined with AWS serverless and microservices architecture - driving up to 40% response times and 14% revenue growth through modernization of monolithic systems to cloud-native platforms. I am skilled in architecture governance, RFP leadership, and translating business strategy into technical execution. Regularly mentors and upskills engineering teams of 20+ on cloud-native and GenAI engineering practices; author of internal engineering standards adopted org-wide.

CORE COMPETENCIES

- Solution & Enterprise Architecture:** Cloud-native architecture, microservices, solution design, fit-gap analysis, architecture governance, RFP leadership
- GenAI / Agentic AI:** RAG, LangChain, LangGraph, Model Context Protocol (MCP), AWS Bedrock, Claude on Bedrock, prompt engineering, embeddings, vector databases, chunking & preprocessing
- Cloud (AWS):** EC2, ECS, EKS, Lambda, S3, RDS, DynamoDB, Bedrock, API Gateway, EventBridge, SQS/SNS, IAM, KMS
- Banking & Domain:** Core Banking (T24, FLEXCUBE), Digital/Neo Banking, Payments, AML, CRM, BFSI, Insurance, Retail, Telecom
- Backend & Languages:** Java, Python, Spring / Spring Boot, Struts, JavaScript, REST/SOAP APIs, JSON/XML
- Leadership & Delivery:** SDLC ownership, technical roadmap, capacity & sprint planning, stakeholder management, cross-functional leadership, talent development, mentorship (20+ engineers)
- Tools:** Sparx Enterprise Architect, PlantUML, Draw.io, JIRA, Confluence, Git/GitLab/GitHub, Jenkins, SonarQube, Figma

EMPLOYMENT HISTORY

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|---|----------------------------------|---------------------------------|
| • Nagarro | Associate Distinguished Engineer | (September 2021 - Present) |
| • Mobiquity | Principal Engineer | (July 2017 - August 2021) |
| • Citicorp Services India Limited | Manager (Tech Lead) | (June 2015 - July 2017) |
| • Tieto Software Technologies Ltd. | Lead Software Engineer | (December 2009 - May 2015) |
| • Clarice Technologies Pvt. Ltd. | Senior Software Engineer | (November 2008 - December 2009) |
| • Spartan Labs Pvt. Ltd. | Software Engineer | (August 2007 - October 2008) |
| • Inbitech Solutions | Software Engineer | (May 2006 - August 2007) |

RECENT PROJECTS (PRESENT -> 2015)

Agentic Ops Platform - Leading Healthcare & Pharma Client

Solution Architect | December 2025 – Present | Python, RAG, LangChain, LangGraph, AWS Lambda, DynamoDB, S3, PostgreSQL (pgvector)

Cloud-native operations automation platform converting SOP-driven business processes into structured, executable task configurations for AI agents.

- Mentored a cross-functional team of 20+ (developers, BAs, testers) on solution design and delivery best practices, improving code quality and reducing rework
- Designed backend architecture spanning API Gateway, AWS Lambda, DynamoDB, and S3; built client, project, SOP-upload, connector, and dashboard modules
- Built SOP parsing, validation, and step-normalization services transforming free-text SOPs into structured, agent-ready configurations
- Implemented RAG-based chatbot and knowledge management; integrated AWS Bedrock for normalizing free-text agent steps into structured JSON tasks
- Designed the Agentic AI orchestration layer using LangChain/LangGraph patterns for agent configuration and execution readiness

Digital Insurance Platform - Leading Life Insurer, South Africa

Solution Architect (SPOC) | May 2023 – November 2025 | AWS, Java, Microservices, Flutter, Camunda, T24, SWIFT

Enhanced digital capabilities with new insurance products, secure online premium payments, and WhatsApp-based automated customer notifications; drove policy persistency through targeted business conservation.

- Led end-to-end technical delivery as SPOC for a team of 15+, owning solution design, client relationships, and delivery governance
- Architected WhatsApp notification workflows for Online Payments, Business Conservation, and Outbound Credit Control
- Delivered Bulk Quoting, Credit Control Reporting, and Business Conservation allocation capabilities
- Provided technical coordination for the Home Loan Protector initiative and authored production support documentation

National Home Loan Platform - Leading Bank, UAE

Solution Architect | June 2022 – April 2023 | AWS, Java, Microservices, Flutter, Camunda, T24, SWIFT

- Architected end-to-end digital home-loan onboarding and KYC journey via mobile application, integrated with core banking, risk, and credit systems
- Delivered backend microservices and cloud-native integration on AWS; led fit-gap analysis across Core Banking (T24) and Payments (SWIFT)

Middleware Modernization - Leading Bank, UAE

Solution Architect | October 2021 – May 2022 | AWS, Python, Kubernetes, Kong Gateway, T24, SWIFT

- Assessed and modernized core banking middleware from monolith to microservices/API-led architecture; authored the cloud migration blueprint
- Derived API rate limits, timeouts, and performance benchmarks for banking workloads integrated with Core Banking (T24) and SWIFT

Neo Banking Platform - Leading Bank, Bahrain

Solution Architect | January 2021 – August 2021 | Java, Microservices, AWS, Backbase, iOS, Android, Kubernetes, Helm

- Delivered an omnichannel banking app covering onboarding, KYC, CASA account opening, and core retail banking, integrated with Core Banking (FLEXCUBE)
- Defined backend architecture for payments, cards, transfers, and digital wallet features; secured IAM integration via Keycloak

Kerbside Pickup - Major US Convenience Store & Fuel Retailer

Solution Architect | March 2020 – December 2020 | Java, Microservices, AWS, iOS, Android

- Designed backend architecture and led technical delivery for a contactless, no-contact mobile shopping and pickup experience

Fuel Pay - Major US Convenience Store & Fuel Retailer

Solution Architect | January 2019 – February 2020 | Java, Microservices, AWS, iOS, Android, Google Analytics

- Designed backend architecture for a contactless mobile fuel payment app, integrating third-party services and the P97 payment gateway

Event Management System - Global Cloud Computing Platform Provider

Solution Architect | July 2017 – December 2018 | Node.js, AWS, Ionic, Cordova

- Designed backend architecture and led technical delivery for a conference event-discovery, booking, and travel-planning platform

Cross-Border Trade Management - Leading Bank (Global Trade Finance Division)

Tech Lead | June 2015 – July 2017 | Java, Oracle, PL/SQL, WebSphere

- Led the technical team delivering an end-to-end global trade-processing platform spanning importer-to-exporter workflows across all trade products

EARLY CAREER (2006 – 2015)

Software Engineer to Lead Software Engineer at Tieto Software Technologies, Clarice Technologies, Spartan Labs, and Inbitech Solutions - delivering enterprise Java/J2EE applications across Core Banking, Trade Finance, and Enterprise Middleware platforms (Oracle, WebSphere, JBoss) for banking clients, building the technical foundation for architecture and delivery leadership. Full project details available on request.

CERTIFICATIONS

- AWS Certified Solutions Architect – Associate

EDUCATION

- Bachelor of Engineering – Information Technology, Bharati Vidyapeeth's College of Engineering, Kolhapur, Maharashtra, India - 2006